IDEA League

Master of Science in Applied Geophysics
Research Thesis

From Particles to Waves

Towards coupling Dynamic Snow Modelling and Spectral Elements to detect Avalanche Breakoff in DAS Data.

Salome Anna Bachmann

Thursday July 9, 2026

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Master of Science Thesis

for the degree of Master of Science in Applied Geophysics
by

Salome Anna Bachmann

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IDEA LEAGUE
JOINT MASTER'S IN APPLIED GEOPHYSICS

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Abstract

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Chapter 1

Introduction

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities in Switzerland, which arecaused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significhant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain for potential avalanches. Furthermore, detecting slab breakoff with distributed acoustic sensing is also important to further understand crack dynamics in snow as well as the influences on crack propagation, such as snowpack properties.

In this master's thesis, the aim is to create an accurate spectral element model (SE) for crack propagation in snow. This is done by pure spectral element simulations, but also by translating material point method simulations into SE. This helps towards understanding the processes associated with crack propagation in snow, that eventually lead to slab breakoff.

In this thesis, a fibre-optic cable installed on a slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (Trabattoni et al., 2023). Strain is the first spatial derivative of deformation, so deformation is the integral in respect to space of strain.

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied. The same is true for the seismic signatures associated with crack propagation leading to slab breakoff. This is important because the time between initial crack propagation and slab breakoff is crucial for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

A multitude of models have been previously used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Brun et al., 1992; Bartelt and Lehning, 2002). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient or water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Brun et al., 1992; Bartelt and Lehning, 2002). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and the consequences they have for the structure of snow. While it is important to understand the influences of for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Furthermore, SNOWPACK and CROCUS do not model the seismic signal itself, which is what is needed to detect and understand crack propagation in DAS. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the disrete element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall and Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall and Strack, 1979). This method has been used by Bobillier et al., 2020; Bobillier et al., 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall and Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

The same is true for finite element modelling (FE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model wave propagation in snow since the 1970s, a review of the different approaches has been published by Podolskiy et al., 2013. It is important to note thet the FE apporach used in this thesis is concerned with wave propagation in snow and not the mechanics of snow itself, so it is fundamentally different from other methods. The finite element modelling approach used in this thesis is described more in detail in ??. FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate Lian et al., 2012. This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (Zhang et al., 2023). Furthermore, MPM does not include boundary conditions in the way FE does (Coombs et al., 2025). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (Zienkiewicz et al., 2025). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (C. A. Knight and N. C. Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models such as Mulak and Gaume, 2019; Gaume et al., 2019 also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl and WeiSSgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl and WeiSSgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl and WeiSSgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and WeiSSgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Trottet et al., 2022; Bobillier et al., 2024; Bergfeld et al., 2025). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet et al., 2022).These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Trottet et al., 2022; Bergfeld et al., 2025). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2018; Gaume et al., 2017; Gaume et al., 2015). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer,meaning the lower its Young's Modulus, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak and Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak and Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak and Gaume, 2019).

Because each of the variables described such as weak layer thickness, critical crack length and so on, alter the velocity, geometry, and energy release of the failure in the snowpack, they directly control the characteristics of the resulting seismic signature. In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasiev et al., 2017).

The different snow processes described have different implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seimsic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

Chapter 2

Theory

2.1 Spectral Element Wave Propagation Modeling

The model described in ?? is implemented using the finite-element (FE) method in SALVUS. The following section describes the physics used by SALVUS. Understanding the phsysics behind the modeling has relevance for understanding and interpreting the results, but also what can be modeled. To discretize a domain, the domain is divided into a finite number of elements (Zienkiewicz et al., 2025). The number of elements in this case is defined by the maximum wave frequency that needs to be resolved accurately, the number of mesh elements per wavelength is user-input. The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (Zienkiewicz et al., 2025). The governing equations are determined at every element node, so for rectangular elements with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and $u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement (Zienkiewicz et al., 2025). The residual or characteristic relationship of linear elastic elements is of the form $r^1 = f^1 - K^1 u^1$, where f are the nodal forces distributing load on the nodes and K is the stiffness matrix of a specific element (Zienkiewicz et al., 2025). In FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (Afanasiev et al., 2019). This means integrals can be calculated over surfaces or even volumes (Afanasiev et al., 2019).

Snow is a porous medium (Mellor, 1977), which is approximated as a solid in this Thesis. This is because SALVUS meshes need an isotropic, solid material and cannot account for pore spaces. Higher frequencies in the resulting data would be atteunated much quicker in an actual porous medium (Koushik et al., 2024). In addition, the presence of pore spaces in the medium would lead to complex scattering of the fields in the results, their absence is an important factor to consider when comparing simulated data to real-world data. The presence of air or water in snow is not neglected completely in the SALVUS simulations implemented however, because the material properties taken from Guillemot et al., 2021 are based on field measurements of snow. This means that the velocities measured by Guillemot et al., 2021 are not of pure snow, but snow that contains air or water.

The governing equation considered in this case is the elastic wave equation, because real-life crack propagation emits P- and S-waves. The wave equation for a heterogenous elastic solid in the n-dimensional domain G is

$$\rho \partial_t^2 u = \nabla \cdot \sigma + f \tag{2.1}$$

(Afanasiev et al., 2019; Aki and Richards, 1981). Stress can be related to displacement using Hoek's law: $\sigma = \frac{C}{\nabla u}$, where C is a 4th order tensor which characterizes the stiffness of the medium (Afanasiev et al., 2019). Combining ?? with Hook's law, this leads to

$$\rho \partial_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \tag{2.2}$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\partial G} = \bar{\tau}(x), u|_{t=0} = \bar{u}(x), v|_{t=0} = \bar{v}(x)$ (Afanasiev et al., 2019). In this case, $\bar{\tau}(x), \bar{u}(x), \bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (Afanasiev et al., 2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\partial G}$.

FE methods use the weak form of the governing equations (Zienkiewicz et al., 2025). To obtain the weak form of the linear elastic wave equation in ??, the whole equation needs to be multiplied with a test function w, which is not time-dependent (Afanasiev et al., 2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{2.3}$$

(Afanasiev et al., 2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) dV = \oiint_s (F \cdot \hat{n}) dS$, where F is the vector field. When this is applied to the first term on the right in ??, this leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_{\partial G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{2.4}$$

(Afanasiev et al., 2019). The first integral on the left side of ?? is the boundary integral, which changes depending on the boundary conditions implemented. The inegral on the right hand side is the mass term, the second on the left hand side is the stiffness term, and the last on the left hand side is the external forcing term, all in ??. Equation ?? can be compacted to

$$(\rho \partial_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\partial G} + (\sigma, \nabla w)_G = (f, w)_G, \tag{2.5}$$

using the notations $(a, b)_G = \int_g a \cdot b d^n x, \langle a, b \rangle_{\partial_G} = \int_{\partial G} a \cdot b d^{n-1} x$ as well as Hoek's law (Afanasiev et al., 2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M,B,S,F\}} I_I(P) = 0 \tag{2.6}$$

(Afanasiev et al., 2019).

To solve ?? the Galerkin approach is used, the test function w and solution u are approximated by ϕ, which are a set of functions from the same space (Afanasiev et al., 2019). Any discrete function, for example displacement can be approximated this way:

$$g(x) \approx \sum_{j=1}^{N} F_j \phi_j(x), \tag{2.7}$$

where F_j is the expansion of the basis and g is the continous function which is the approximation (Afanasiev et al., 2019). The derivatives of a function can then be approximated by the derivatives of ϕ.

To evaluate the integrals in ??, numerical quadrature is used Afanasiev et al., 2019. A certain number of intergration weights W are combined, g(x) and the derivative of g(x) are evaluated at a set of integration points x_i (Afanasiev et al., 2019). This resluts in

$$\int_G g(x)d^n x \approx \sum_{i=1}^{M} W_i^G \sum_{j=1}^{N} F_j \phi_j(x_i) \tag{2.8}$$

(Afanasiev et al., 2019). In FE, an element-specific reference coordinate system θ is used. In this thesis, the domain consists of a snow layer and an air layer, which means there is a sharp discontinuity in model parameters. As Afanasiev et al., 2019 state, the model parameters and strain field are not differentiable. This is why a smooth polynomial basis of the governing equations over the whole domain is not appropriate, the basis B needs to consist of piecewise functions G defined on the individual elements (Afanasiev et al., 2019). The integration weights and basis function derivatives used, according to Afanasiev et al., 2019, are pre-tabulated in the element-specific coordinate system θ. To transfrom the integrals and derivaties to physical coordinates x, so the coordinates of the snow extent in this case, the element-specific Jacobian $\frac{\delta x}{\delta \theta}$ is used (Afanasiev et al., 2019). Equations ?? and ?? and their derivatives become:

$$g(x) \approx \sum_{N}^{j=1} F_j \phi_j(\theta(x)) \tag{2.9}$$

and

$$\nabla g(x) \approx \sum_{j=1}^{N} F_j \frac{\partial}{\partial \theta} \phi_j(\theta(x)) \frac{\partial \theta}{\partial x} \tag{2.10}$$

(Afanasiev et al., 2019). In equations ?? and ??, the function space V is approximated as $V_h \approx V$ in a previous step, this means the accuracy of the approximation is dependent on how well the original function space is approximated (Afanasiev et al., 2019). To make calculations over the whole domain G, the contributions from each element need to be summed:

$$\int_{G_e} g(x)\, \mathrm{d}^n x \approx \sum_{i=1}^{M} W_i^{G_e} \sum_{j=1}^{N} F_j\, \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^{n} \tag{2.11}$$

for the volume and

$$\int_{\partial G_e} g(x)\, \mathrm{d}^{n-1} x \approx \sum_{i=1}^{M} W_i^{\partial G_e} \sum_{j=1}^{N} F_j\, \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^{n-1} \tag{2.12}$$

for the surface of the domain (Afanasiev et al., 2019).

The dependence on the reference coordinate system is described by E, which is a model containing the physical coordinate to reference coordinate transformation, in other words, the determinant of the Jacobian matrix and its inverse (Afanasiev et al., 2019). Using basis B of the polynomial functions of the approximation V_h, which are piecewise per-element because of possible discontinuities, the abstract form in ?? becomes

$$\sum_{P \in \{M,B,S,F\}} I_I(P, B(E)) = 0 \tag{2.13}$$

(Afanasiev et al., 2019). This means that fields such as displacement and velocity are calculated in this way. From these fields, the data from the model in section ?? is obtained and plotted.

2.2 From Particles to Waves: Translating MPM to SE

The material point simulations of the PST performed by (Gaume et al., 2018; Trottet et al., 2022) result in the recorded stresses in the model according to Gaume et al., 2019. This results in the stretch factor ϵ comprised of the material properties C_{ij} such as Young's modulus, Lamé parameters and Poisson ratio as well as the stress matrix σ:

$$\epsilon = C_{ij}\sigma \tag{2.14}$$

(Holzapfel, 2010). The equation can be rewritten to obtain the stresses in different directions, in the xx direction for example:

$$\sigma_{xx} = (\lambda + 2\mu)\epsilon_{xx} + \lambda\epsilon_{yy} \tag{2.15}$$

(Aki and Richards, 1981). When scaling this with the particle volume, the individual moment tensor components can be obtained using

$$M_{ij} = \sigma_{ij} V_p. \tag{2.16}$$

These moment tensor components can then be input into SALVUS as custom source time functions. How are the waveforms of the stfs obtained

2.3 Simulation of DAS data

To compare the results obtained in the SALVUS simulations to field DAS data, the numerical outputs have to be transformed. The backscattered delay of a laser pulse along a straigth cable is proportional to the integrated strain tensor components along the gauge length (noeFrontiersFullwaveformSeismology2025). This yields the standard expression for the gauge-averaged DAS strain at a channel position:

$$\bar{\epsilon}_{\mathrm{DAS}}(s_c, t) = \frac{1}{l} \int_{s_c - \frac{l}{2}}^{s_c + \frac{l}{2}} \mathrm{e}^T(s)\boldsymbol{\epsilon}(s,t)\mathrm{e}(s)\, ds, \tag{2.17}$$

(noeFrontiersFullwaveformSeismology2025).

To model a physically realistic laser pulse with a tapered spatial intensity profile, the classical box-car averaging window is generalized to a continuous, weighted spatial convolution according to noeFrontiersFullwaveformSeismology2025:

$$\bar{\epsilon}_{\mathrm{DAS}}(x, t) = \int_{-\infty}^{\infty} w(s) \cdot \epsilon_{xx}(x - s, t)\, ds, \tag{2.18}$$

where $w(s)$ represents the normalized Gaussian spatial weighting kernel configured via the Pyber engine (noeFrontiersFullwaveformSeismology2025, located on github), satisfying the amplitude-preserving condition:

$$\int_{-\infty}^{\infty} w(s)\, ds = 1 \tag{2.19}$$

(noeFrontiersFullwaveformSeismology2025).

On the discrete numerical grid output by SALVUS, this forward problem maps to a discrete 1D spatial convolution over N_g sampling points within the gauge length window:

$$\bar{\epsilon}_{\mathrm{DAS}}(x_i, t) \approx \sum_{k=-N_g/2}^{N_g/2} w_k \cdot \epsilon_{xx}(x_{i-k}, t). \tag{2.20}$$

Because spatial convolution and spatial differentiation are both linear operations, they strictly commute. This linear commutativity is exploited numerically to apply the pulse filtering directly to the raw horizontal displacement field u_x prior to differentiation:

$$\bar{\epsilon}_{\mathrm{DAS}}(x_i, t) = \sum_k w_k \cdot \left(\frac{\partial u_x(x_{i-k}, t)}{\partial x} \right) = \frac{\partial}{\partial x} \left[\sum_k w_k \cdot u_x(x_{i-k}, t) \right]. \tag{2.21}$$

Finally, the simulated DAS strain rate is acquired by evaluating the temporal derivative of the stabilized strain field:

$$\dot{\bar{\epsilon}}_{\mathrm{DAS}}(x_i, t) = \frac{\partial}{\partial t} \left[\bar{\epsilon}_{\mathrm{DAS}}(x_i, t) \right]. \tag{2.22}$$

Chapter 3

Methodology

3.1 Model Setups

3.1.1 Simple Model

The same two-layer medium as in section ?? was set up. This medium spans 400m in width and 3m in height. In the simple case, a series of SALVUS moment tensor sources are set up with a certain distance, they span from 30m to 270m with a spacing of 10cm. The model domain setup can be seen in figure MAKE FIGURE.The time delay with which the sources fire is dependend on the sub-rayleigh or supershear case which is implemented, the bounds for the crack propagation velocity are taken from Trottet et al., 2022, they are 0.6 or 1.6 times the shear wave velocity for sub-rayleigh and supershear cases respectively.

The bedrock or soil is not considered in both models. This is because preliminary tests of a 3-layered medium have shown that it acts as a reflector of waves. This is because of the high impedance contrast between snow and bedrock, which causes waves to be reflected almost entirely at the interface Aki and Richards, 1981.For computational efficiency, the bedrock layer was thus neglected and the bottom of the snow was made a regular boundary, in other words, reflecting, to mimick the presence of a layer below the snow. The top as well as both side boundaries of the model are set to be absorbing boundaries, which means that waves that reach these boundaries are absorbed and not reflected back into the model. This is done so that the model accurately represents the physical behavior of wave propagation in a much larger domain, which is the reality of field DAS data. The absorbing boundaries were set to be 3.5 times the maximum wavelength present in the model, this is according to the SALVUS documentation (Afanasiev et al., 2017).

3.1.2 Cohesive Zone Model

A graphical representation of the model setup and the equations used can be found in appendix ??.

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened. This means that before a point in the material cracks, there is already weakening of the material before. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

The mesh is generated to resolve frequencies up to the 95th percentile of the source wavelet's frequency content. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\partial}{\partial_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}}\,(1-\xi) \quad \text{for } 0 \le \xi \le 1, \qquad T(\xi) = 0 \text{ for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{crack}}$ for a crack front with initial position x_0 and local process length l_{crack}.

The mixed-mode potential used in the code follows the form from Mahajan and Joshi, 2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right)\left\{\left[1+\frac{\Delta_n}{\delta_n}\right] + q\left[1-\exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right]\right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e\,\sigma_{\max}\,\delta_n, \qquad \phi_t = \sqrt{\tfrac{e}{2}}\,\tau_{\max}\,\delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1-r_{\text{mode}})\,\delta_{n,c}^2 + r_{\text{mode}}\,\delta_{s,c}^2}\ .$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé parameters: $E = \dfrac{\mu(3\lambda+2\mu)}{\lambda+\mu}$ (Aki and Richards, 1981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1-r_{\text{mode}})\,\delta_{n,c}^2 + r_{\text{mode}}\,\delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from Mahajan and Joshi, 2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{\text{eff}} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in Trottet et al., 2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{\text{start}}}{v_{\text{rupture}}},$$

using the subfault horizontal position x, the rupture starting position x_{start}, and the prescribed rupture speed v_{rupture}.

Each activated subfault is represented as a point source in the SALVUS source array. The kinematic moment formulation serves as the numerical link to the underlying cohesive zone model through a strict energy balance. Within the CZM, the total strain energy released during fracture is split between the fracture energy (G_I or G_{II}) required to break down the cohesive zone, and the remaining radiated elastic strain energy. By prescribing x_{slip} across a finite grid cell width Δx, the resulting scalar seismic moment M_0 mathematically captures the radiated energy that propagates away from the process zone into the bulk medium as a seismic wavefield.

To take the mixed-mode mechanics of the process zone into account, the individual components of the moment tensor are scaled by the mode-mix ratio r_{mode}, which can be user input:

$$m_{xx} = 0, \quad m_{xy} = M_0 \, r_{\text{mode}}, \quad m_{yy} = -(1 - r_{\text{mode}}) M_0.$$

Consequently, at the extremes of $r_{\text{mode}} = 0$ or 1, the unaligned component of the moment tensor drops to zero, perfectly matching the radiation patterns of pure opening or pure shear failure. All subfault sources are driven by an identical central frequency, source wavelet, and amplitude scaling factor, thereby reconstructing the discrete subfault array into a highly coherent, continuously moving source that mimics a smooth, physical crack tip rupture throughout the wave propagation mesh.

Key parameters and their provenance used in the simulations are summarized in appendix ??.

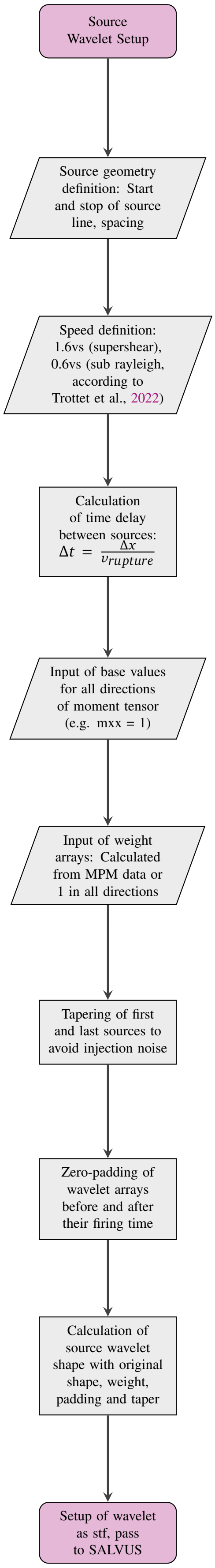
3.2 Source Generation

All sources generated use SALVUS' custom source time function option. For the simple model and the cohesive zone model, this is done by writing a file with the source wavelets in them. The source wavelets have ricker shape according to RickerWavelet2021:

$$ricker_{base} = (1.0 - 2.0 * (\pi * f_0 * t_{local})^2) e^{-((\pi * f_0 * t_local)^2)}.$$

(3.1)

In the MPM case, the shape of the wavelets is obtained as described in ??. The workflow of the code, including how the source speed is set up can be seen in the following flowchart.



Chapter 4

Results

4.1 Simple Model Results

The following results shown are for the supershear and sub-rayleigh case of the custom stfs. The sources are spaced 0.1m apart between 30m and 270m and have a ricker wavelet all directions. The moment tensors have components 1 in xx and xy and -1 in yy.

4.1.1 Sub-Rayleigh Velocity plots

The following plots show horizontal (vx) and vertical (vy) particle velocity at two reciever depths: 2.25m, which is the snow bottom, and 2.625m which is the depth of the crack. The particle velocity shown is derived from the displacement field, it is not the instantaneous particle velocity because instantaneous velocity is not measured in the field.

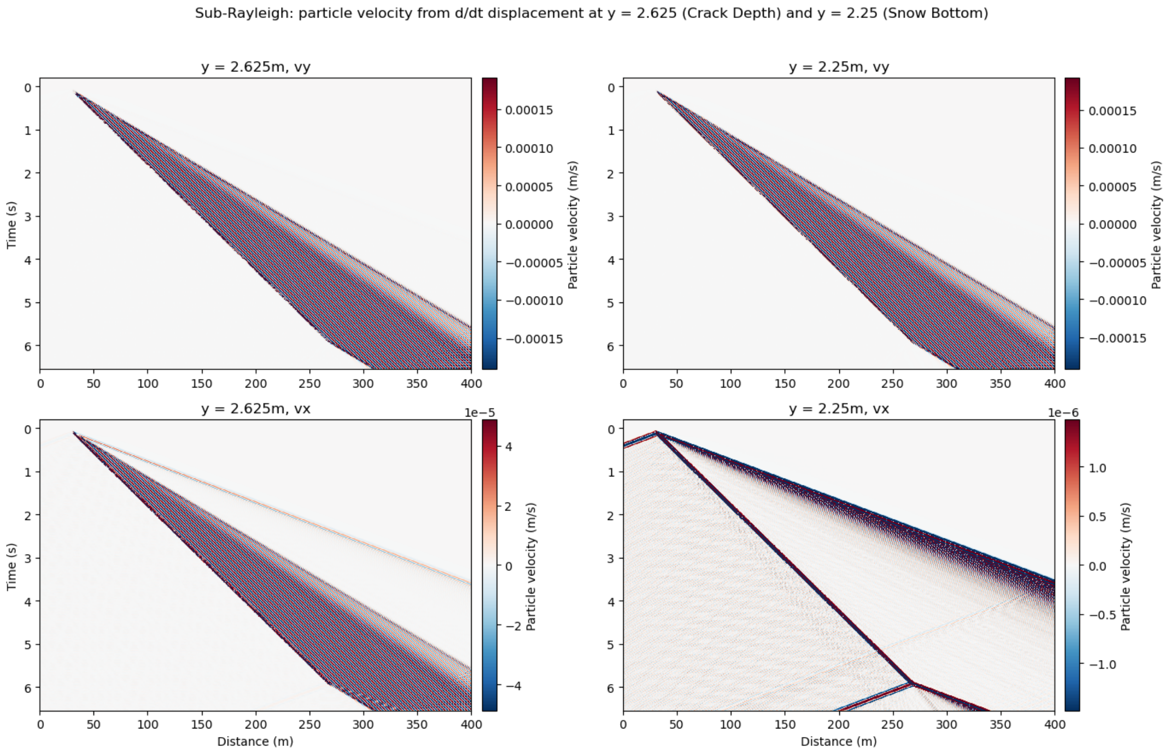


Figure 4.1: Sub-Rayleigh Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

The sub-rayleigh velocity plots in figure ?? all show a distinct wavefront starting at 0s at 30m and ending at 270m. This main wavefront ends at 6s, but sends out waves with a different speed (different angle) in direction of the source propagation. The horizontal velocity at the snow bottom also shows a backward-propagating wavefront from the crack tip.

Both vx plots (bottom row in figure ??), show the main wavefront with waves travelling ahead of it, as time and distance progresses, the main wavefront gets wider. In all but the plot of vx at the snow bottom, the first couple of waves in the main wavefront are weaker in color, and thus weaker in velocity.

In the main wavefront observed, it's very noticeable that the velocity is layered. In other words, there are stripes along the wavefront with opposite polarity, for example, there is a stripe with positive polarity (red) followed by one of negative ploarity (blue). This is also noticeable in the cases that have multiple wavefronts, although it's less pronounced.

In the horizontal velocity plot at the snow bottom but also at crack depth, there are three wavefronts preceding the main wavefront: negative polarity (blue) followed by positive (red) and then negative again. At the snow bottom, this front can be observed twice, one time also coming from the left boundary.

4.1.2 Supershear Velocity Plots

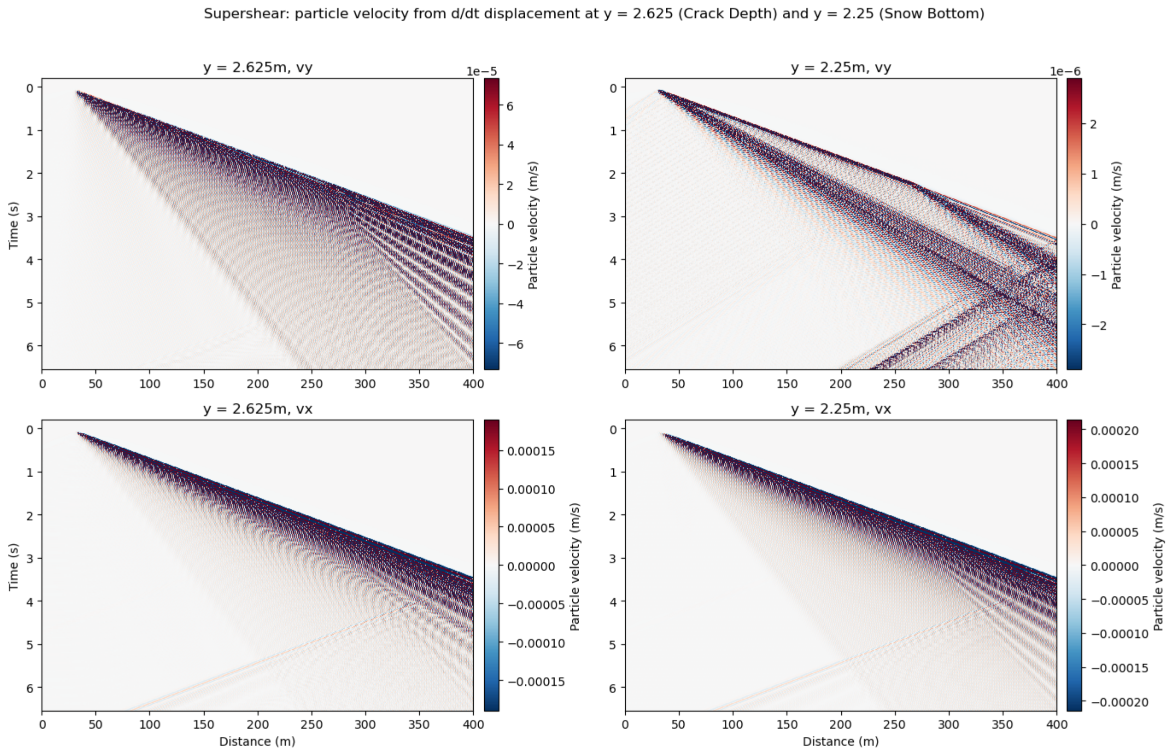


Figure 4.2: Supershear Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

In all of the supershear velocity plots in figure ??, there is a distinct wavefront from 30m to 270m, which also broadens. The wavefront is much broader in the vy plots (top row of figure ??), it is broadest in the vertical velocity case at the snow bottom.

The wavefronts arriving in all cases switch between polarities. While the pattern of polarity is the same in all cases, the slope of the wavefronts change. In all plots, after the first wavefront, the waves seem to curve downward. The vy plots in the top row also show a fan of waves between 2.5s and 3s from 300 to 350m, which radiates from the main wavefront.

4.1.3 Sub-rayleigh strain and strain rate plot

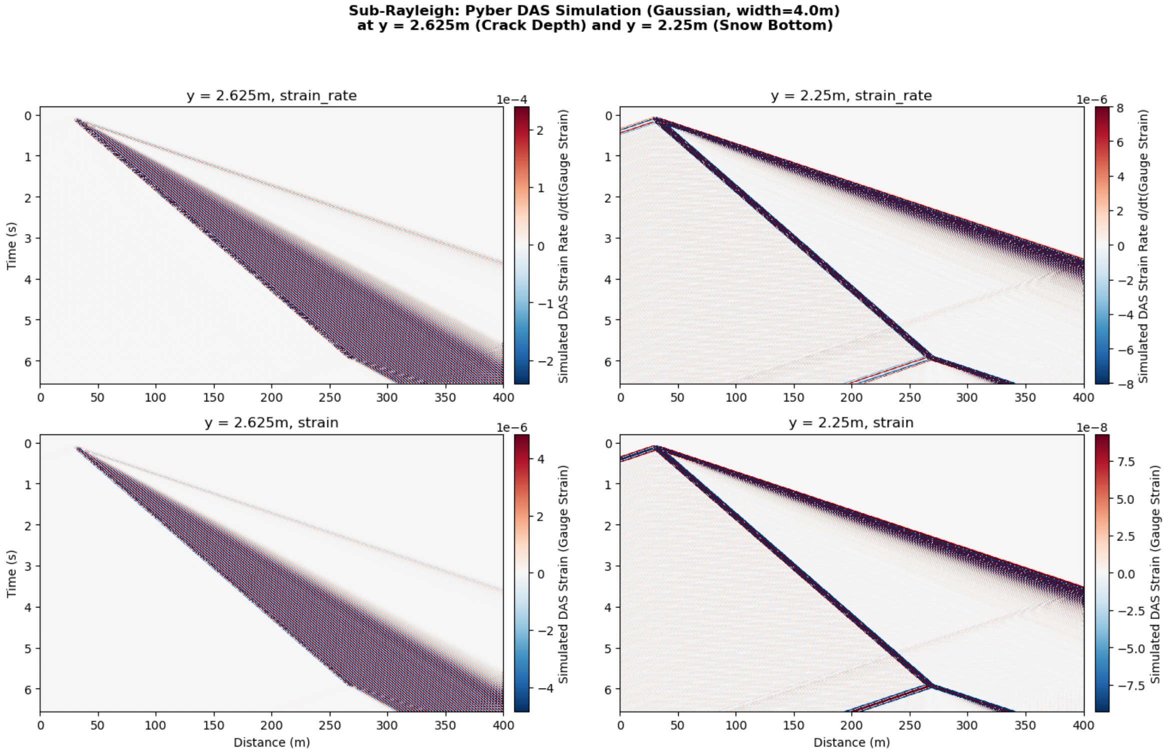


Figure 4.3: Sub-rayleigh strain and strain rate (horizontal component) averaged over 4m gaugue length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

Both the strain and the strain rate at crack depth look very similar to each other, the same is true for the snow bottom. At crack depth, both the strain and strain rate in the left column of figure ?? show one strong main front, which is preceded by a preliminary front. The same is true for the snow bottomb, but the main front is at a different location compared to crack depth.

At crack depth, the main front broadens with increasing time. This is also true for the snow bottom, but here, the broading is much more drastic. Between the individual fronts, there is a change in slope.

In all plots, the strains and strain rates following each other have opposite polarity. The magnitude of strain and strain rate is larger at the crack depth compared to the bottom of the snow.

4.1.4 Supershear strain and strain rate plot

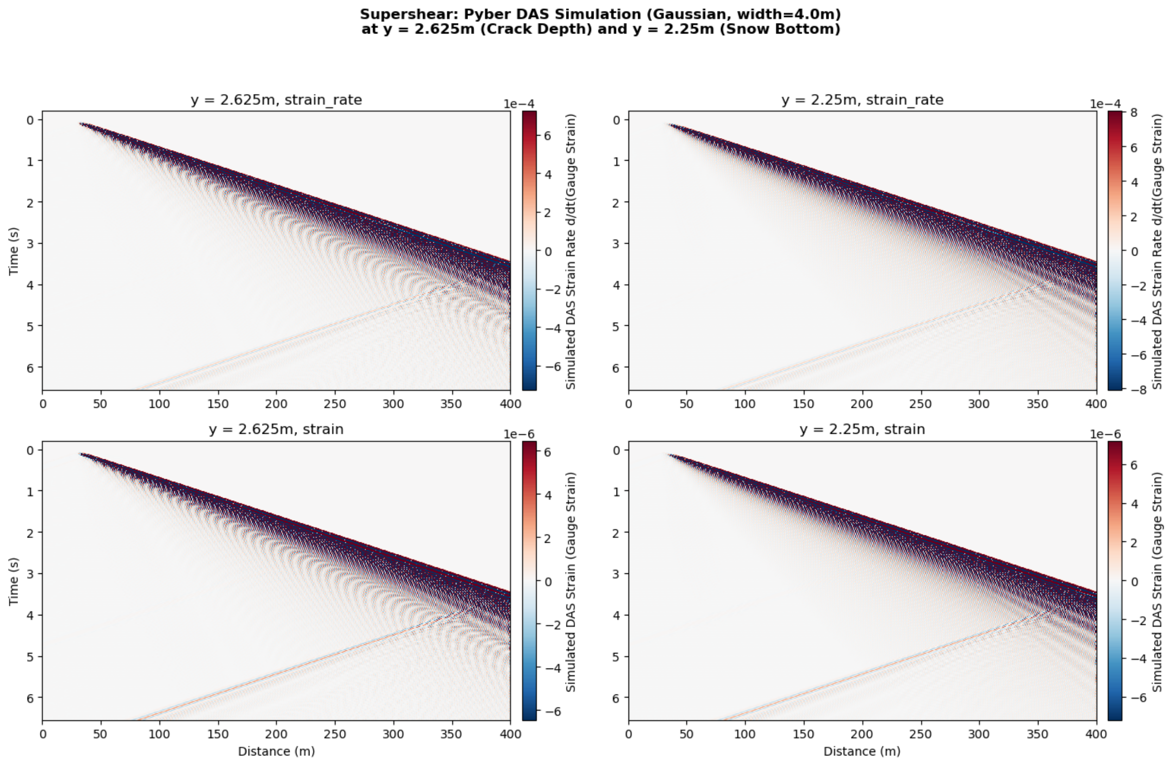


Figure 4.4: Supershear strain and strain rate (horizontal component) averaged over 4m gaugue length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

In the supershear case, strain and strain rate at the snow bottom and crack depth shown in figure ?? all look very similar. There is a high magnitude wavefront arriving first, after which magnitude decays.

The strain and strain rates observed have opposite polarity (positive followed by negative). In the strain rate plots, the different polarities can be seen a bit more clearly than in the strain plots in the bottom row of figure ??.

In figure ??, there is no difference in the magnitude of strain and strain rate between the crack depth and snow bottom.

4.1.5 Frequency-Wavenumber (fk) Plot

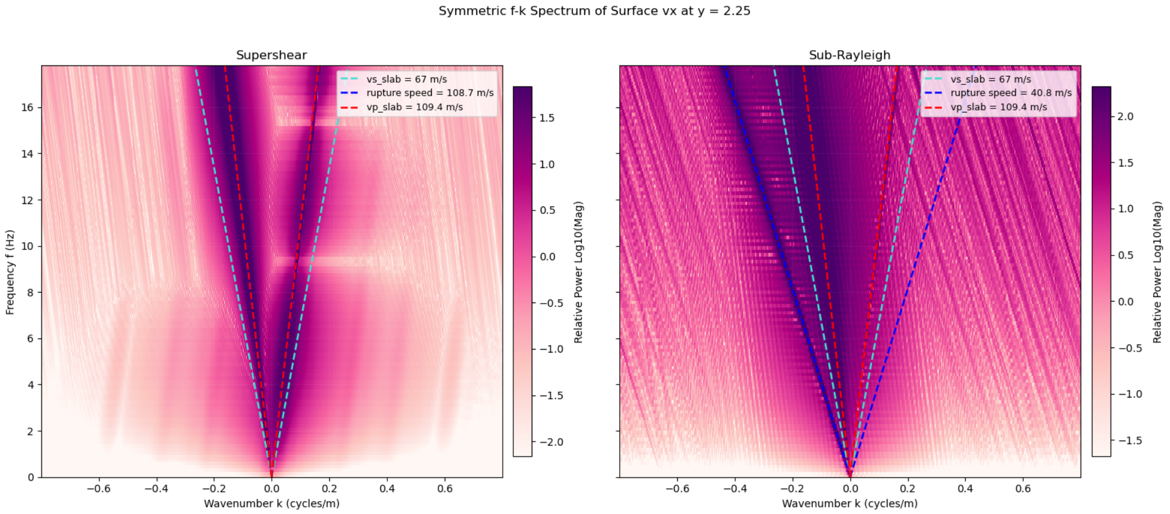


Figure 4.5: F-k plot for the supershear (left) and sub-rayleigh (right) case. Slab p-wave velocity in green, s-wave velocity in black and respective rupture speed in blue. Colorbar shows relative energy (log10 of the magnitide of energy).

When comparing the supershear and the sub-rayleigh fk-plots in figure ??, it becomes clear that the energy is disributed differently in the two cases. The supershear case shows a concentration of energy between the rupture speed (blue line) and the vp (green line) of the slab on both sides of the spectrum. The sub-rayleigh case on the other hand shows three concentrations of energy, two around the positive and negative vp of the slab and one at the negative rupture speed. In all cases, the energy is more concentrated around the vp and or rupture speed for lower frequencies, at higher frequencies, the concentration of energy starts to deviate from speeds present in the medium in figure ??.

In the supershear case, the dark spots of concentrated energy are more narrow on the positive side compared to the negative side, where they are broader and more smeared out. In the sub-rayleigh case on the right side of figure ??, the dark lines of concentated energy are much more narrow and distinct.

Figure ?? also clearly shows the distinction in where the energy concentrates. In the sub-rayleigh case, as described above, the energy is concentrated around the p-wave speed on both sides and the rupture speed only in negative direction. In supershear crack propagation, the energy is concentrated between vp and the rupture speed.

Chapter 5

Discussion

5.1 Simple Model

5.1.1 Velocity space-time plots

The main wavefronts which can be observed in figures ?? and ?? are the crack fronts. This is because when taking the slope of these two wavefronts, it corresponds to the rupture velocity in the supershear and the sub-rayleigh case.

The weakening of the wavefronts, especially observed in the supershear case in figure ?? can be explained of attenuation of the different frequencies which are present in the medium. Higher frequencies are attenuated much faster than low frequencies (Aki and Richards, 1981). The attenuation coefficient is

$$\alpha(\omega) = \frac{\omega}{2cQ} \tag{5.1}$$

(Aki and Richards, 1981), which can be used to define the spatial amplitude decay:

$$A(x, \omega) = A_0 e^{-\alpha(\omega)x} = A_0 e^{-\frac{\omega x}{2cQ}} \tag{5.2}$$

(Aki and Richards, 1981). This means that higher frequencies present in the wave train decay in amplitude faster than others with increasing distance. The resulting wavefield is described by the equation

$$u(x, t) = \int_{-\infty}^{\infty} U_0(\omega) e^{-\frac{\omega x}{2cQ}} e^{i\omega\left(\frac{x}{c} - t\right)} d\omega \tag{5.3}$$

(Aki and Richards, 1981). This is only present in the supershear case, because the crack travels faster than the guided waves (Dunham and Archuleta, 2005). A Mach-front is created, where all the waves travel together in a "shock-front" (Dunham and Archuleta, 2005), but once this front passes, there is attenuation of energy. In the sub-rayleigh case on the other hand, the guided waves are faster than the crack (Aki and Richards, 1981), there is reflection of the waves between the top and bottom of the snow, which creates the osicllations which can be seen in figure ??.

The broadening of the wavefront with increasing time can be explained by dispersion. By the reflections of the top and bottom snow boundary, the waves present start interfering with each other, which can lead to constructive or destructive interference (Aki and Richards, 1981). This means that waves can start travelling at different phase velocities, because of the interference Graff, 1975. The phase and group velocities of waves are described as

$$v_{\text{ph}} = \frac{\omega}{k}, \quad v_{\text{g}} = \frac{d\omega}{dk}. \tag{5.4}$$

The interference leads to the broadening of the wavefront with increasing time, described by the envelope spreading law (Aki and Richards, 1981)

$$\Delta x(t) = \Delta x_0 \sqrt{1 + \left(\frac{\frac{d^2\omega}{dk^2} t}{(\Delta x_0)^2} \right)^2}. \tag{5.5}$$

As time goes on, there are more reflections present, so there is more potential for interference and thus dispersion.

The presence of waves with opposite polarity can be proven mathematically. From the equation of motion in an elastic solid, which is

$$\mu \nabla^2 u + (\lambda + \mu) \nabla (\nabla \cdot u) = \rho \frac{\partial^2 u}{\partial t^2}, \tag{5.6}$$

the scalar and vector potential for P- and S-waves (Potentials ϕ, ψ)can be derived using the Helmholtz-Decomposition and the displacement becomes

$$u = \nabla \phi + \nabla \times \psi. \tag{5.7}$$

Splitting this into the horizontal and vertical component of displacement:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \tag{5.8}$$

and

$$u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x}. \tag{5.9}$$

When these are substituted into ??, it becomes

$$\nabla^2 \phi = \frac{1}{c_p^2} \frac{\partial^2 \phi}{\partial t^2} \tag{5.10}$$

for P-waves and

$$\nabla^2 \psi = \frac{1}{c_s^2} \frac{\partial^2 \psi}{\partial t^2} \tag{5.11}$$

for S-waves, the wave speeds are derived from the Lamé-Parameters, as described in ??.

When assuming harmonic plane wave solutions for waves with wavenumber k and frequency ω, the harmonic solutions have the form

$$\phi(x, z, t) = f(z) e^{i(kx - \omega t)} \tag{5.12}$$

and

$$\psi(x, z, t) = g(z) e^{i(kx - \omega t)} \tag{5.13}$$

. When these are substituted into the wave equations, f(z) and g(z) yield ordinary differental equations, which are

$$\frac{d^2 f}{dz^2} + p^2 f = 0 \tag{5.14}$$

and

$$\frac{d^2 g}{dz^2} + q^2 g = 0, \tag{5.15}$$

with $p^2 = \frac{\omega^2}{c_p^s} - k^2$ and $q^2 = \frac{\omega^2}{c_s^s} - k^2$. These equations lead to the general solutions

$$\phi = [Acos(pz) + Bsin(pz)]e^{i(kx-\omega t)} \tag{5.16}$$

and

$$\psi = [Ccos(qz) + Dsin(qz)]e^{i(kx-\omega t)}. \tag{5.17}$$

At the snow-air interface, there is a stress-free boundary because of the impedance contrast, accoding to Igel, 2016. This means that both the normal and the shear stress need to be zero:

$$\sigma_{zz} = \lambda\nabla^2\phi + 2\mu\frac{\partial u_z}{\partial z} = 0 \tag{5.18}$$

and

$$\sigma_{xz} = \mu(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}) = 0. \tag{5.19}$$

These can be expressed in terms of the scalar and vector potential using the displacement potentials.

$$\sigma_{zz} = \lambda(\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial z^2}) + 2\mu(\frac{\partial^2\phi}{\partial z^2} + \frac{\partial^2\psi}{\partial x\partial z}), \tag{5.20}$$

$$\sigma_{xz} = \mu(\frac{\partial^2\phi}{\partial x\partial z} - \frac{\partial^2\psi}{\partial z^2} + \frac{\partial^2\phi}{\partial x\partial z} + \frac{\partial^2\psi}{\partial x^2}), \tag{5.21}$$

of which the first terms can be substituted with the wavespeed using the wavenumber-frequency and velocity relationship from the Lamé parameters. This results in

$$\sigma_{zz} = \mu[(q^2 - k^2)\phi + \frac{\partial^2\psi}{\partial x\partial z}] \tag{5.22}$$

and

$$\sigma_{xz} = \mu[2\frac{\partial^2\phi}{\partial x\partial z} + \frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi}{\partial z^2}], \tag{5.23}$$

from which harmonic solutions containing sin and cos can be obtained. Here, a derivative is a multiplication with ik, which leads to

$$\sigma_{zz} = \mu[(q^2 - k^2)\phi + 2ik\frac{\partial\psi}{\partial z}] \tag{5.24}$$

and

$$\sigma_{xz} = \mu[2ik\frac{\partial\phi}{\partial z} + (q^2 - k^2)\psi]. \tag{5.25}$$

When taking into account the stress free boundary and substituting $K = (q^2 - k^2)$, then the normal stress at z becomes

$$\sigma_n = K[Acos(pz) + Bsin(pz) + 2ik(-q(sin(qz) + qDcos(pz)))] = 0 \tag{5.26}$$

and for the shear stress

$$\sigma_s = 2ik[-pAsin(pz) + pBcos(pz)] + K[Ccos(qz) + Dsin(qz)] = 0 \tag{5.27}$$

at the top and botto boundary of the snow layer. The matrix for symmetric Lamb waves is obtained by adding the top and bottom term of the normal stress and subtracting the shear stresses. This leads to

$$2KAcos(pz) + 4ikqDcos(qz) = \tag{5.28}$$

and

$$-4ikpAsin(pz) + 2KDsin(qz) = \tag{5.29}$$

, which can be written as a matrix:

$$\begin{pmatrix} Kcos(pz) & 2ikqcos(qz) \\ -2ikpsin(pz) & Ksin(qz) \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{5.30}$$

If the determinant of this matrix is set to zero, this results in

$$\frac{tan(qz)}{tan(pz)} = -\frac{2k^2pq}{(q^2 - k^2)^2} \tag{5.31}$$

, which is the dispersion of the symmetric mode of the Lamb waves, as described by Su and Ye, 2009. The same can be done for the asymmetric Lamb wave mode if the operations of the stresses are opposite.

The presence of opposite polarities across both velocity components (v_x, v_y) and across both propagation regimes (sub-Rayleigh and supershear) is a direct consequence of the waveguide's geometric symmetry and the nature of the fracture source. Dynamically, the two recording depths, $y = 2.625$ m and $y = 2.25$ m, are perfectly symmetric about the mid-plane of the snow slab ($y = 2.4375$ m), mapping precisely to $\pm z$ in the analytical framework.

The fact that this phase-reversal persists across both the sub-Rayleigh and supershear regimes demonstrates that while the rupture velocity changes the radiation pattern (e.g., generating dispersive wave trains versus non-dispersive Mach shock fronts), the underlying phase relationships remain strictly governed by the structural eigenfunctions of the plate waveguide.

The presence of opposite polarities in both the horizontal and vertical velocity fields simultaneously indicates that the wavefield is not a single mode, but rather a superposition of both symmetric (u^{sym}) and antisymmetric (u^{asym}) modes:

$$u_x(x, \pm z, t) = u_x^{\mathrm{sym}}(x, \pm z, t) + u_x^{\mathrm{asym}}(x, \pm z, t) \tag{5.32}$$

$$u_z(x, \pm z, t) = u_z^{\mathrm{sym}}(x, \pm z, t) + u_z^{\mathrm{asym}}(x, \pm z, t) \tag{5.33}$$

Because a crack propagating along a weak layer acts as an asymmetric boundary source, it breaks the symmetry of the slab and couples energy into both structural types simultaneously. Evaluating Equations ?? and ?? at $+z$ versus $-z$ activates the opposing parities of the underlying sin and cos depth functions:

$$u_x(x, -z, t) = u_x^{\mathrm{sym}}(x, z, t) - u_x^{\mathrm{asym}}(x, z, t) \tag{5.34}$$

$$u_z(x, -z, t) = -u_z^{\mathrm{sym}}(x, z, t) + u_z^{\mathrm{asym}}(x, z, t) \tag{5.35}$$

This modal interference shifts the cumulative phase balances, causing a reversal of wave-train polarity between the crack depth and the snowpack bottom for both components of motion (Su and Ye, 2009).

Lamb waves are elastic guided waves that occur in plates (Su and Ye, 2009). They consist of particle motion in the direction of the wave polarization as well as the direction perpendicular to it (Su and Ye, 2009). They are guided by the boundaries of the layer or media in which they propagate and have a symmetric and an asymmetric mode (Su and Ye, 2009). The symmetric mode dominates the in-plate motion, while the antisymmetric mode dominates the vertical (out of plane motion) (Su and Ye, 2009).

5.1.2 Strain and strain rate space-time plots

The sub-rayleigh horizontal strain and strain rate plots in figure ?? look very similar to the velocity plots in figure ??. The same is true for the supershear strain and strain rate in figure ??, which look similar to all but the vertical velocity at the snow bottom (top right in figure ??). This is intuitive because strain is the spatial derivative of displacement ($\frac{\partial u}{\partial x}$), strain rate is the temporal derivative of strain ($\frac{\partial^2 u}{\partial x \partial t}$). Velocity on the other hand is the spatial derivative of displacement ($\frac{\partial u}{\partial t}$), so all the values are derived from displacement.

Both the sub-rayleigh and the supershear strain and strain rate cases show a wave train running ahead. The slope of the first wavefront observed in figures ?? correspond to the p-wave velocity in snow and the rupture velocity in the supershear case. This means that in both cases, the p-waves are the first arrivals because they are fastest. This is consistent with the observations made in the velocity plots. In the supershear case, the p-waves are followed immediately by the waves from the rupture front, which would explain why there is only one clear but very broad wavefront in figure ??. In the sub-rayleigh strain and strain rate plots in figure ??, the observations are also consistent with velocity. The first wave front to arrive are the p-waves, followed by the front from the rupture and the s-waves last. The same goes for the widening of the wavefronts, which has already been discussed in the velocity section.

The first arrival of the p-wave is also consistent with wave propagation. All plots in figures ?? show positive horizontal strain or strain rate first, followed by compression (negative strain) and positive strain again. This propagation is characterized by extension first and then contraction (Aki and Richards, 1981).

This very repetitive sequence of opposite polarities is a consequence of the snow acting as a waveguide, as described above. When energy is suddenly released by a propagating crack tip, the crack acts as an impulsive source that projects disturbances into the overlying slab (Heierli et al., 2008; Gaume et al., 2018). The leading P-wave arrival corresponds to the high-frequency, long-wavelength limit of the symmetric (S_0) Lamb mode, which represents an in-plane, extensional push-pull wave packet (Achenbach, 1973; Graff, 1975). As this transient S_0 pulse outruns the rupture front, the initial forward displacement stretches the material, yielding a leading zone of positive tensile strain (red color). To satisfy elastodynamic restoring forces and conserve momentum as the wavefront passes a fixed point in space, this extension must be immediately followed by a compensating contractional phase of negative compressive strain (blue), before returning to equilibrium.

Behind this fast, extensional precursor lies the slower, highly oscillatory wave train, which displays a dense pattern of alternating opposite polarities. This trailing pattern is dictated by the superposition of the symmetric mode as well as the antisymmetric (A_0) flexural Lamb wave mode. Similar to the dispersion of the symmetric mode of the Lamb waves in equation ??, the antisymmetric wave mode can be derived. For the antisymmetric mode, the vertical displacement must be an even function of depth, $u_z(-z) = u_z(z)$, while the horizontal displacement must be an odd function, $u_x(-z) = -u_x(z)$. To satisfy the geometric properties, the integration constants representing the even component of ϕ and the odd component of ψ must vanish, such that $A = 0$ and $D = 0$. The asymmetric potentials simplify to:

$$\phi = B \sin(pz) e^{i(kx-\omega t)} \tag{5.36}$$

$$\psi = C \cos(qz) e^{i(kx-\omega t)}. \tag{5.37}$$

Substituting Equations ?? and ?? into the stress-free boundary equations ($\sigma_{zz} = 0, \sigma_{xz} = 0$) at the free surfaces $\pm z$ yields the following homogeneous matrix system:

$$\begin{pmatrix} K\sin(pz) & -2ikq\sin(qz) \\ 2ikp\cos(pz) & K\cos(qz) \end{pmatrix} \begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{5.38}$$

where $K = (q^2 - k^2)$. Setting the determinant of the matrix to zero gives the characteristic dispersion relation for the antisymmetric Lamb wave modes:

$$\frac{\tan(pz)}{\tan(qz)} = \frac{4k^2pq}{(q^2 - k^2)^2} \tag{5.39}$$

To relate this explicitly to the strain, the horizontal displacement u_x is computed from the potentials:

$$u_x = \frac{\partial\phi}{\partial x} - \frac{\partial\psi}{\partial z} = [ikB\sin(pz) + qC\sin(qz)]\, e^{i(kx-\omega t)} \tag{5.40}$$

Taking the spatial derivative with respect to x to find the horizontal strain ($\varepsilon_{xx} = \partial u_x/\partial x$) transforms the expression via multiplication by ik in the wavenumber domain:

$$\varepsilon_{xx}^{\text{asym}}(x, z, t) = \left[-k^2B\sin(pz) + ikqC\sin(qz)\right] e^{i(kx-\omega t)} \tag{5.41}$$

By isolating the real part of this analytical expression, the spatial and temporal variation of the flexural strain field isolates the reason for the opposite polarity patterns along the wave train. The strain can be rewritten in terms of its harmonic phase:

$$\varepsilon_{xx}^{\text{asym}}(x, z, t) = \mathcal{A}_{\text{asym}}(z)\cos(kx - \omega t + \phi_0) \tag{5.42}$$

where $\mathcal{A}_{\text{asym}}(z)$ represents the depth-dependent amplitude envelope and ϕ_0 is a constant phase angle. The transcendental cosine term dictates that over a fixed spatial distance of exactly half a wavelength, $\Delta x = \lambda/2 = \pi/k$, the wavefield must cycle smoothly between its maximum positive value ($\cos = +1$) and its maximum negative value ($\cos = -1$).

Physically, this represents the dynamic bending or structural "ringing" of the slab (Heierli et al., 2008; Bergfeld et al., 2025). When a section of the slab flexes vertically, the fibers located on a single side of the neutral axis experience localized horizontal squeezing ($\varepsilon_{xx} < 0$, blue bands). As the elastic wave oscillates into its next half-cycle, the local curvature reverses, subjecting those same fibers to intense horizontal stretching ($\varepsilon_{xx} > 0$, red bands).

When this is translated to the horizontal strain rate ($\dot{\varepsilon}_{xx}$), the time derivative ($\partial/\partial t \longleftrightarrow -i\omega$) shifts the wave profile into a sine function:

$$\dot{\varepsilon}_{xx}^{\text{asym}}(x, z, t) = \omega\mathcal{A}_{\text{asym}}(z)\sin(kx - \omega t + \phi_0) \tag{5.43}$$

This differentiation scales the amplitude by the angular frequency ω and introduces a 90° spatial phase shift, but leaves the underlying alternating pattern of opposite polarities completely intact. Because both the crack depth ($y = 2.625$ m) and the snowpack base ($y = 2.25$ m) reside entirely within the lower half of the slab waveguide below its neutral axis ($z < 0$), they evaluate to the same geometric sign in $\sin(pz)$. Consequently, both recording horizons register these rapid, cyclic transitions from extension to contraction perfectly in-phase with one another.

5.1.3 Discussion of fk-plot

The narrower energy band on the positive side of the spectrum, as observed in the supershear case in figure ?? can be explained by source directivity. Ben-Menahem, 1961 derive directionality factors for seismic waves propagating on either side of a fault. The directionality factor is

$$D = \left| \frac{\left(\frac{C}{v} + \cos\theta_0\right) \sin\left[\frac{\pi b}{\lambda}\left(\frac{C}{v} - \cos\theta_0\right)\right]}{\left(\frac{C}{v} - \cos\theta_0\right) \sin\left[\frac{\pi b}{\lambda}\left(\frac{C}{v} + \cos\theta_0\right)\right]} \right| \tag{5.44}$$

(Ben-Menahem, 1961). Analogous to the case described here, where the crack starts at one point in the medium, where the energy travels in the direction of the crack but also opposite to the direction of crack propagation. Ben-Menahem, 1961 describe that energy is concentrated in the direction of propagation, so in the positive direction. When looking at the directivity function in equation ?? at 0 degrees (direction of propagation), the function becomes

$$D = \left| \frac{\left(\frac{C}{v} + 1\right) \sin\left[\frac{\pi b}{\lambda}\left(\frac{C}{v} - 1\right)\right]}{\left(\frac{C}{v} - 1\right) \sin\left[\frac{\pi b}{\lambda}\left(\frac{C}{v} + 1\right)\right]} \right| \tag{5.45}$$

, so if the phase velocity (C) is close to the rupture velocity (v), this can lead to a concentration of energy. The opposite is true in the negative direction (180 degrees), where the amplitude is reduced. This explains why the energy is more concentrated and less smeared out in the positive direction compared to the negative direction.

The concentration between vp and the rupture speed in the supershear case can be explained by the radiation of energy. The rupture velocity in the medium exceeds the shear wave speed, so the s-waves no longer decay, they radiate away from the fault as Mach cones (Dunham and Archuleta, 2005). These waves have a certain angle which is described by

$$\xi = \sin^{-1}\left(\frac{c_s}{v}\right) = \tan^{-1}\left(\frac{1}{\hat{\eta}_s}\right) \tag{5.46}$$

, so is determined by the ratio between the rupture speed and the s-wave speed (Dunham and Archuleta, 2005). These waves are tied to the rupture front, which can explain why the energy is concentrated between the p-wave speed and the rupture speed in the supershear case in figure ??. The p-wave speed in this case is still faster than the rupture, so this energy outruns the rupture, which means the p-waves precede. The rupture and the Mach cone associated with it arrive second, which explains why the energy is concentrated between the two, according to Rosakis, 2002.

In the sub-rayleigh case, the energy is concentated at the p-wave speeds, which are the fastest speeds present in the medium. The energy is dominated by the starting and stopping phases, which are radiated by the abrupt starting and stopping of the rupture (Madariaga, 1977). These are governed by the material speeds, so the p- and s-wave speeds (Madariaga, 1977). The energy concentration observed at the negative rupture speed could be explained by evanescence of the waves (Dunham and Archuleta, 2005), which means that the amplitude of the waves decays exponentially with their distance to the fault. The furthest distance from the end of the fault is opposite to the direction of propagation, which means that the p- and s-wave energy has decayed the most there. This could allow energy from the fault to show up in the fk plot in figure ?? at that location because it is no longer overshadowed by the stronger energy of the propagating waves. This could also explain why this energy is only observed in the negative direction.

The strong contrast in how the energy is concentrated between the sub-rayleigh and supershear case in ?? can be explained by the number of locations the energy is concetrated. As described above, the supershear regime has two energy fronts, whereas the sub-rayleigh case has only one. This explains why the energy bands in the supershear case are much more broad compared to the sub-rayleigh case.

The observation that at higher frequencies in figure ??, the lines of the energy present do not match as well anymore with the concentration of energy, can be explained by numerical dispersion. As Komatitsch and Tromp, 1999 describe, as frequency increases, the wavelength decreases. This means that higher frequencies are not resloved as well by a spatial grid with a certain spacing Yilmaz, 2001, this starts bending the energy in the fk plot. This could explain the mismatch between the physical velocity lines and the concetrations of energy.

Chapter 6

Outlook

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Chapter 7

Conclusion

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The subfault onset time is given by $\boldsymbol{t_o}$.

Horizontal coordinates match $\boldsymbol{x}$.

Appendix A

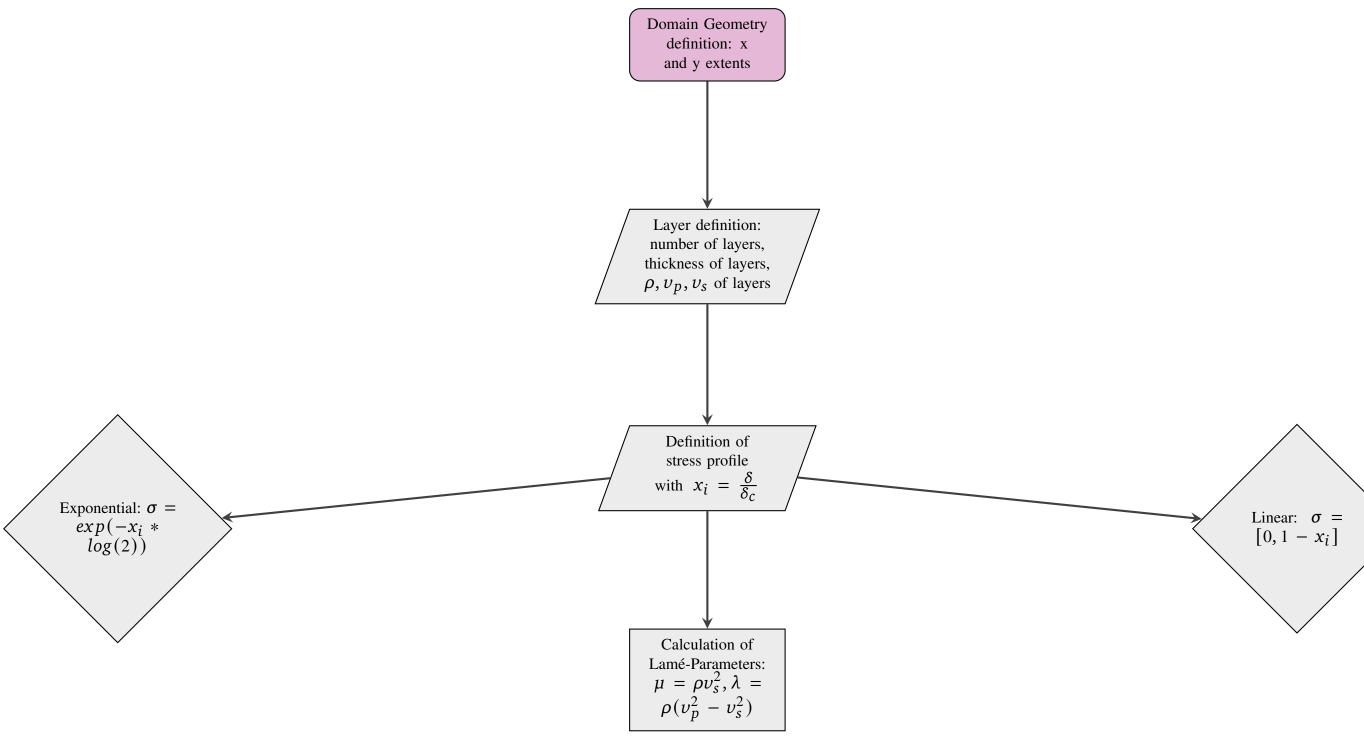
Code Repository

The complete code of all models described can be found on github: https://github.com/s-bachmann/Master-Thesis.

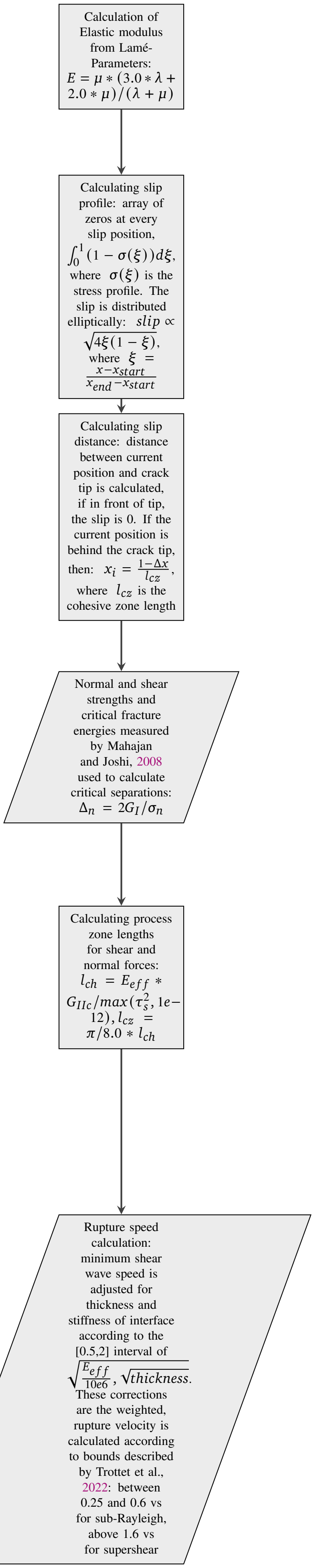
Appendix B

Code Workflow

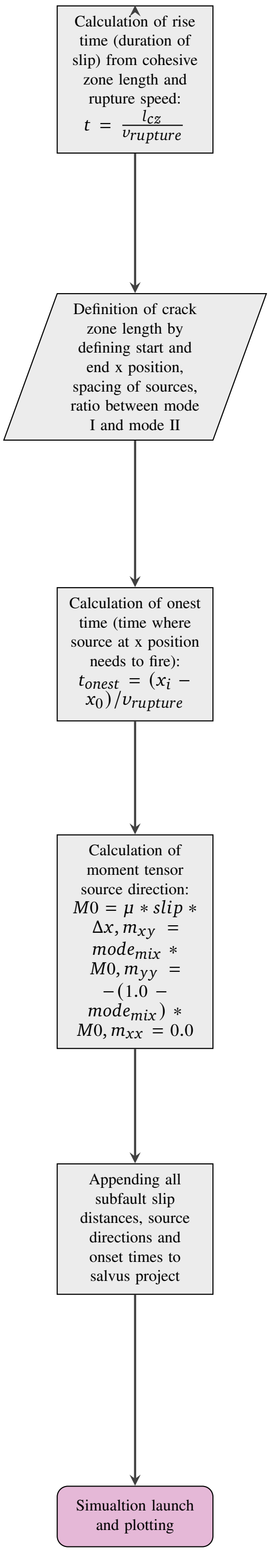
Part 1: Domain and Stress Profile Setup



Part 2: Material Properties and Process Zone Calculation



Part 3: Rupture Propagation and Source Definition



Appendix C

Parameters used in Models

Parameter	Typical value / source
σ_n	0.00184 MPa from Mahajan and Joshi, 2008
τ_s	0.004 MPa from Mahajan and Joshi, 2008
G_I, G_{II}	0.35, 0.50 $J/m^s$from Mahajan and Joshi, 2008
v_s, v_p, ρ	67.9366, 109.4 m/s, 250 kg/mş from Trottet et al., 2022 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

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